

Regime-Adaptive Algorithmic Trading: A Comprehensive Synthesis of Econometrics, Deep Learning, and Hybrid Architectures

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Abstract—The domain of quantitative finance has undergone a radical transformation, shifting from rigid, linear econometric models to flexible, data-driven Artificial Intelligence (AI) architectures. This paper presents a comprehensive research study that bridges the gap between theoretical foundations, systematic literature review, and practical implementation of advanced trading strategies. We begin by establishing a robust theoretical framework that covers classical econometrics (ARIMA, GARCH), modern Deep Learning (LSTM, Transformers), and Reinforcement Learning agents.

We then proceed to a Systematic Literature Review (SLR) covering 2020-2025, identifying the emergence of "Hybrid Architectures" as a dominant paradigm. Building on these insights, we propose and rigorously test a modular algorithmic trading framework comparing three distinct approaches: (1) A Volatility Targeting mechanism utilizing a Hybrid GARCH-LSTM model to dynamically adjust leverage, (2) An Adaptive Bollinger Band strategy driven by GARCH volatility forecasts, and (3) A Momentum Gatekeeper utilizing XGBoost to filter trade entries.

Experimental results on out-of-sample data (2025-2026) across a diverse universe of assets (SPY, AAPL, NVDA, TSLA, etc.) reveal distinct regime dependencies. The Adaptive Bollinger strategy achieves a Sharpe Ratio > 1.7 on mean-reverting assets like SPY, while the XGBoost Gatekeeper demonstrates superior capital preservation in bear markets, achieving a Sharpe Ratio > 2.0 on GOOG. This study concludes that while AI models offer superior pattern recognition, their success is contingent upon regime-specific application and robust risk management.

Index Terms—Algorithmic Trading, Quantitative Finance, Hybrid Architectures, GARCH-LSTM, XGBoost, Deep Reinforcement Learning, Volatility Targeting, Regime Switching.

I. INTRODUCTION

Financial markets are complex adaptive systems characterized by non-stationarity, high noise-to-signal ratios (SNR), and "regime-y" behavior. Structural breaks primarily driven by macroeconomic events—such as pandemics, interest rate shifts, and geopolitical conflicts—render static models obsolete. Historically, quantitative finance relied on linear stochastic models like the Autoregressive Integrated Moving Average (ARIMA) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH). While these models provide statistical rigor and interpretability, they fundamentally assume linear dependencies and often fail to capture the chaotic, non-linear nature of high-frequency market data.

The advent of Deep Learning (DL) has introduced a new paradigm. Neural networks, particularly Recurrent Neural Networks (RNNs) like Long Short-Term Memory (LSTM) units, offer the capacity to approximate any continuous function, allowing them to model complex non-linear patterns. However, standard DL models suffer from opacity ("black-box" nature) and a tendency to overfit noisy financial data.

This paper argues for a "Hybrid" approach. Rather than discarding econometrics for deep learning, or vice versa, we explore the synergy between the two. By using econometrics to model the "known" linear structure (e.g., volatility clustering via GARCH) and Deep Learning to model the "unknown" non-linear residuals, we can achieve superior predictive performance and robustness.

A. Research Objectives

The primary objectives of this study are:

- 1) **Theoretical Unification:** To map the landscape of modern AI in finance, providing rigorous mathematical definitions for key algorithms (GARCH, LSTM, XGBoost, RL).
- 2) **Systematic Review:** To synthesize state-of-the-art research from 2020 to 2025, identifying trends in Hybrid Architectures and Natural Language Processing (NLP).
- 3) **Empirical Validation:** To implement and backtest three distinct strategies (Volatility Targeting, Adaptive Mean Reversion, Momentum Gatekeeping) to evaluate their performance across different market regimes.

B. Key Contributions

This paper makes the following contributions:

- A detailed mathematical formulation of a **Hybrid GARCH-LSTM** model where LSTM predicts the residuals of a GARCH variance forecast.
- An analysis of an **XGBoost-based Gatekeeper** that uses classification probability as a meta-labeler to filter false breakouts.
- A comprehensive experimental evaluation using a **Walk-Forward** methodology on 12 major assets, utilizing the **Garman-Klass** estimator for robust volatility handling.

- A discussion on the "Cost of Insurance," highlighting how risk-averse strategies underperform in unilateral bull markets (e.g., NVDA).

II. THEORETICAL BACKGROUND

This section establishes the mathematical foundations for the models used in this research. We traverse from classical econometrics to modern machine learning.

A. Econometric Foundations

1) ARIMA (Autoregressive Integrated Moving Average):

ARIMA is the cornerstone of linear time-series forecasting. It decomposes a time series into three components:

- **AR(p)**: The current value depends linearly on its own previous p values.
- **I(d)**: Differencing the series d times to achieve stationarity.
- **MA(q)**: The current value depends linearly on current and past error terms.

Mathematically:

$$(1 - \sum_{i=1}^p \phi_i L^i)(1 - L)^d X_t = (1 + \sum_{j=1}^q \theta_j L^j) \epsilon_t \quad (1)$$

where L is the lag operator. While effective for trend following, ARIMA assumes constant variance (homoskedasticity), a violation of financial reality.

2) **GARCH (Generalized Autoregressive Conditional Heteroskedasticity)**: Financial returns exhibit "volatility clustering"—large changes tend to be followed by large changes. GARCH(p, q), introduced by Bollerslev [1], models the conditional variance σ_t^2 as a function of past squared residuals and past variances.

The standard GARCH(1,1) implies fixed lag orders. In our implementation, however, we determine the optimal orders (p, q) for each asset via Grid Search, selecting the model that minimizes the Akaike Information Criterion (AIC). The general GARCH(p, q) process is given by:

$$\sigma_t^2 = \omega + \sum_{i=1}^p \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2 \quad (2)$$

Constraints: $\omega > 0, \alpha_i \geq 0, \beta_j \geq 0$, and $\sum \alpha_i + \sum \beta_j < 1$ for stationarity. This model captures the persistence of volatility shocks, making it crucial for risk management.

3) **Maximum Likelihood Estimation (MLE)**: The parameters $\Theta = (\omega, \alpha, \beta)$ are estimated by maximizing the log-likelihood function. Assuming the residuals ϵ_t follow a normal distribution $\epsilon_t \sim N(0, \sigma_t^2)$, the log-likelihood is:

$$\mathcal{L}(\Theta) = -\frac{T}{2} \log(2\pi) - \frac{1}{2} \sum_{t=1}^T \left(\log(\sigma_t^2) + \frac{\epsilon_t^2}{\sigma_t^2} \right) \quad (3)$$

In practice, financial returns often exhibit "fat tails" (leptokurtosis). Therefore, we employ a Student's t-distribution or Skewed Student's t-distribution for the likelihood function, which adds a degrees-of-freedom parameter ν to the optimization space.

4) **Stationarity and Differencing**: A critical prerequisite for time-series modeling is stationarity. A time series is strictly stationary if the joint distribution of any collection of random variables is invariant to time shifts. Weak stationarity requires constant mean and variance:

- $E[X_t] = \mu$ (Constant Mean)
- $Var(X_t) = \sigma^2$ (Constant Variance)
- $Cov(X_t, X_{t+k}) = \gamma_k$ (Covariance depends only on lag k)

Financial prices P_t are non-stationary (Random Walk). Thus, we model log-returns r_t :

$$r_t = \ln(P_t) - \ln(P_{t-1}) \quad (4)$$

which are typically stationary in mean but not in variance (hence the need for GARCH).

B. Deep Learning Architectures

1) **Long Short-Term Memory (LSTM)**: LSTMs [2] are a specialized RNN architecture designed to mitigate the vanishing gradient problem, enabling the learning of long-term dependencies. The core component is the memory cell c_t , regulated by three gates:

- **Forget Gate (f_t)**: Decides what to discard from the cell state.

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (5)$$

- **Input Gate (i_t)**: Decides what new information to store.

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (6)$$

- **Output Gate (o_t)**: Determines the output based on cell state.

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (7)$$

The cell state is updated as:

$$c_t = f_t * c_{t-1} + i_t * \tanh(W_c \cdot [h_{t-1}, x_t] + b_c) \quad (8)$$

In our Hybrid framework, the LSTM is not trained on raw prices, but on the **residuals** of the GARCH model, effectively learning the non-linear error structure.

2) **Backpropagation Through Time (BPTT)**: Training an LSTM involves unrolling the network across time steps and applying the chain rule. The error gradients are propagated backward from the output at $t = T$ to the initial state at $t = 0$. This process, known as Backpropagation Through Time (BPTT), is computationally intensive but allows the network to learn temporal dependencies that span hundreds of time steps. The loss function for regression tasks is typically the Mean Squared Error (MSE):

$$\mathcal{L} = \frac{1}{T} \sum_{t=1}^T (\hat{y}_t - y_t)^2 \quad (9)$$

Optimization is performed using Adam (Adaptive Moment Estimation), which adapts the learning rate for each parameter based on the first and second moments of the gradients.

3) *Temporal Fusion Transformers (TFT)*: While LSTMs excel at sequential processing, they struggle with heterogeneous data types and long-range dependencies. The Temporal Fusion Transformer (TFT) [11] introduces an attention-based architecture designed for multi-horizon forecasting. Key innovations include:

- **Gating Mechanisms (GRN)**: Gated Residual Networks skip unused components, allowing the model to adapt its depth.
- **Variable Selection Networks**: Explicitly weight the importance of each input feature, providing interpretability.
- **Temporal Self-Attention**: Captures long-term value dependencies independent of distance.

C. Ensemble Learning: XGBoost

eXtreme Gradient Boosting (XGBoost) [3] is a scalable implementation of gradient boosted decision trees. Unlike Random Forests which build independent trees, XGBoost builds trees sequentially to correct the errors of previous trees.

The objective function consists of a training loss l and a regularization term Ω :

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (10)$$

where $\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ controls complexity (T is number of leaves, w are leaf weights). This regularization makes XGBoost particularly robust against overfitting in noisy financial datasets compared to standard GBMs.

1) *Split Finding Algorithm*: The core innovation of XGBoost is its scalable split finding. For a given tree node, the algorithm calculates the potential "Gain" from splitting a leaf into two children (Left and Right):

$$Gain = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (11)$$

where G and H are the sums of the first and second order gradients of the loss function in the respective leaves. If $Gain < 0$, the split is rejected (pruning). This exact formulation allows XGBoost to optimize any differentiable loss function, including custom financial objectives like Sharpe Ratio maximization.

D. Reinforcement Learning (RL)

Deep Reinforcement Learning (DRL) frames trading as a Markov Decision Process (MDP).

- **State (S_t)**: Market features (Price, Volume, Tech Indicators).
- **Action (A_t)**: Buy, Sell, or Hold.
- **Reward (R_t)**: PnL or Sharpe Ratio.

Key algorithms include:

- **PPO (Proximal Policy Optimization)**: A policy gradient method that clips updates to ensure stability.
- **A2C (Advantage Actor-Critic)**: Uses a critic network to estimate the value function, reducing variance.

While we do not implement a full PPO agent in the practical section due to stability concerns, the Momentum Gatekeeper shares conceptual similarities by learning a policy (When to enter?) based on reward maximization.

E. Performance & Risk Metrics

To evaluate the efficacy of trading strategies, we move beyond simple returns to risk-adjusted metrics.

1) *Sharpe Ratio*: The Sharpe Ratio measures the excess return per unit of deviation risk.

$$SR = \frac{E[R_p - R_f]}{\sigma_p} \quad (12)$$

where R_p is the portfolio return, R_f is the risk-free rate, and σ_p is the standard deviation of the portfolio's excess return. A ratio > 1.0 is generally considered acceptable, while > 2.0 represents excellent risk-adjusted performance.

2) *Sortino Ratio*: The Sortino Ratio improves upon Sharpe by penalizing only downside volatility.

$$SoR = \frac{E[R_p - R_f]}{\sigma_d} \quad (13)$$

where σ_d is the standard deviation of negative asset returns (downside deviation). This is particularly relevant for strategies (like Momentum) that seek upside volatility while minimizing crashes.

3) *Maximum Drawdown (MDD)*: MDD measures the largest peak-to-trough decline in portfolio value.

$$MDD = \max_{t \in [0, T]} \left(\max_{\tau \in [0, t]} (V_\tau) - V_t \right) / \max_{\tau \in [0, t]} (V_\tau) \quad (14)$$

Minimizing MDD is often more critical for institutional investors than maximizing total return, as it dictates capital preservation.

F. Technical Indicators

To support our strategy logic, we employ several standard technical indicators.

1) *Relative Strength Index (RSI)*: The RSI is a momentum oscillator that measures the speed and change of price movements. It is defined as:

$$RSI = 100 - \frac{100}{1 + RS} \quad (15)$$

where RS (Relative Strength) is the ratio of the average gain to the average loss over a specified period. It helps identify overbought or oversold conditions.

2) *Moving Averages (MA)*: We utilize the Simple Moving Average (SMA) for trend identification:

$$SMA_t(n) = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i} \quad (16)$$

Comparing price against long-term SMAs (e.g., 200-day) acts as a robust regime filter.

G. Optimization Framework: Optuna & TPE

For hyperparameter tuning, we utilize Bayesian Optimization via the **Tree-structured Parzen Estimator (TPE)** [8]. Unlike Grid Search, which is exhaustive and inefficient ($O(N^d)$), TPE models the conditional probability $P(x|y)$ rather than $P(y|x)$.

The algorithm maintains two distributions for a parameter x :

$$P(x|y) = \begin{cases} l(x) & \text{if } y < y^* \\ g(x) & \text{if } y \geq y^* \end{cases} \quad (17)$$

where y^* is a quantile of the observed objective values (e.g., the top 15% of scores). $l(x)$ models the density of parameters that yielded "good" results, while $g(x)$ models the "bad" results. The TPE algorithm then selects the next parameter to evaluate by maximizing the Expected Improvement (EI), which is proportional to the ratio $l(x)/g(x)$. This effectively drives the search towards regions where the probability of being in the "good" distribution is high relative to the "bad" distribution.

III. RELATED WORK

A Systematic Literature Review (SLR) of studies published between 2020 and 2025 reveals three dominant themes in the application of AI to finance.

A. Hybrid Architectures

The most robust results often come from combining econometrics with DL. Kashif and Slepaczuk [4] demonstrated that an **ARIMA-LSTM** hybrid significantly outperformed standalone LSTM models on the S&P 500. By using ARIMA to filter the linear trend, the LSTM could focus its capacity on modeling non-linear anomalies. Similarly, Romdhane and Boubaker [9] utilized a **HAR-LSTM-GARCH** model for energy markets, proving that neural networks effectively correct the rigid variance assumptions of classical GARCH.

B. Transformers and NLP

The integration of unstructured data is a major frontier. Ruan and Jiang [5] showed that **FinBERT**-enhanced models achieved Sharpe ratios > 1.5 by incorporating news sentiment. The "Attention" mechanism in Transformers, such as the **Temporal Fusion Transformer (TFT)** used by Lee [11], offers interpretability by weighing the importance of different inputs, addressing the "black box" criticism.

C. Dynamic Optimization

Recent work focuses on optimizing for risk-adjusted returns rather than raw accuracy. Wang and Liu [12] found that **PPO** agents achieved superior Sortino Ratios compared to supervised models by learning to avoid drawdowns. Liu [6] introduced **OW-XGBoost**, which optimizes a multi-objective function (Return, Std Dev, Sharpe), achieving a Sharpe ratio of 3.11 on the CSI 300.

IV. METHODOLOGY

We propose a modular framework designed to test hypothesis across different market regimes.

A. Data Preprocessing: Garman-Klass Volatility

Standard Close-to-Close volatility estimation ignores intra-day price action. We employ the **Garman-Klass** estimator [7], which utilizes Open, High, Low, and Close prices to provide a more efficient estimator of variance:

$$\sigma_{GK}^2 = 0.5 \ln \left(\frac{H_t}{L_t} \right)^2 - (2 \ln 2 - 1) \ln \left(\frac{C_t}{O_t} \right)^2 \quad (18)$$

This serves as the ground truth target for our volatility models.

B. Strategy 1: Volatility Targeting (Hybrid GARCH-LSTM)

This strategy aims to stabilize returns by scaling position size inversely to forecasted risk.

1) *Hybrid Forecasting*: We first fit a GARCH model to the log-returns. We then calculate the residuals:

$$e_t = \sigma_{GK,t} - \hat{\sigma}_{GARCH,t} \quad (19)$$

An LSTM network is trained to predict e_{t+1} using a lookback window of technical features. The final forecast is the sum of the linear and non-linear components:

$$\hat{\sigma}_{Hybrid,t+1} = \hat{\sigma}_{GARCH,t+1} + \text{LSTM}(X_{t-k:t}) \quad (20)$$

2) *Leverage Control*: The target leverage is calculated as:

$$L_t = \frac{\sigma_{target}}{\hat{\sigma}_{Hybrid,t+1}} \quad (21)$$

This is further modulated by a **Trend Filter**: if $P_t < MA_{trend}$, the leverage is multiplied by a scalar γ (Bearish Exposure), effectively de-risking during downtrends.

Algorithm 1 Hybrid GARCH-LSTM Training Pipeline

Require: Log-returns r_t , Realized Volatility $\sigma_{GK,t}$

- 1: Estimate GARCH parameters Θ via MLE
 - 2: Generate linear forecasts $\hat{\sigma}_{GARCH,t} \leftarrow \text{GARCH}(r_{1:t-1}, \Theta)$
 - 3: Calculate residuals $e_t \leftarrow \sigma_{GK,t} - \hat{\sigma}_{GARCH,t}$
 - 4: Normalize technical features X_t
 - 5: Train LSTM to minimize $\mathcal{L} = \sum (\hat{e}_t - e_t)^2$
 - 6: **return** Trained Hybrid Model
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C. Strategy 2: Adaptive Bollinger Bands

Standard Bollinger Bands use a simple Moving Average and constant Standard Deviation. We replace the static SD with the GARCH conditional volatility forecast.

- **Upper Band:** $MA_{window} + k \cdot \hat{\sigma}_{GARCH}$
- **Lower Band:** $MA_{window} - k \cdot \hat{\sigma}_{GARCH}$

The bands dynamically expand during high-volatility regimes (widening the "value" zone) and contract during calm periods. A Long signal is generated only if the price touches the Lower Band AND the long-term trend is bullish (Filter).

D. Strategy 3: Momentum Gatekeeper (XGBoost)

This strategy addresses the "false breakout" problem.

1) *Base Signal*: A simple Moving Average crossover or trend check suggests market entry.

TABLE I: Key Studies in AI-Driven Quantitative Finance (2020-2025)

Author (Year)	Focus	Methodology	Key Insight
Kashif et al. (2025)	Hybrid Forecasting	ARIMA-LSTM	Residual modeling improves Information Ratio by isolating non-linear noise.
Romdhane (2026)	Volatility	HAR-LSTM-GARCH	Deep Learning effectively corrects GARCH variance estimates in energy markets.
Ruan & Jiang (2025)	NLP Sentiment	FinBERT + SHAP	Textual sentiment acts as an orthogonal alpha signal, boasting Sharpe > 1.5 .
Lee (2025)	Crypto Assets	TFT (Transformer)	Attention mechanisms provide interpretability for variable selection.
Wang & Liu (2025)	Portfolio Opt	PPO (DRL)	RL agents optimize Sortino Ratio better than supervised regression models.
Liu (2024)	Multi-Objective	OW-XGBoost	Custom loss functions (Sharpe) outperform standard MSE in trade classification.

Algorithm 2 Adaptive Bollinger Signal Generation

Require: Price P_t , GARCH Volatility $\hat{\sigma}_t$, Trend MA M_L , Band MA M_S

- 1: Calculate Upper Band $U_t \leftarrow M_S + k \cdot \hat{\sigma}_t$
- 2: Calculate Lower Band $L_t \leftarrow M_S - k \cdot \hat{\sigma}_t$
- 3: **if** $P_t > M_L$ **then**
- 4: **if** $P_t < L_t$ **then** **Signal** $\leftarrow$ BUY
- 5: **else if** $P_t > U_t$ **then** **Signal** $\leftarrow$ CLOSE
- 6: **else** ($P_t \leq M_L$)
- 7: **Signal** $\leftarrow$ NEUTRAL (No Longs)
- 8: **end if**
- 9: **return** Signal

2) *The Gatekeeper:* An XGBoost classifier analyzes the market context (RSI, Volatility Ratios, Volume). It outputs a probability score $P(U_p|State)$.

3) *Decision:* The trade is executed only if $P(U_p) > \tau$, where τ is a strict threshold (e.g., 0.6). This transforms the strategy from a "always-in" system to a "sniper" system.

Algorithm 3 Momentum Gatekeeper Logic

Require: Price P_t , Features X_t , Classifier $\mathcal{M}$, Threshold τ

- 1: **Signal** $\leftarrow$ NEUTRAL
- 2: **if** Trend is Bullish ($P_t > MA_{fast}$) **then**
- 3: Calculate Probability $p \leftarrow \mathcal{M.predict_proba}(X_t)$
- 4: **if** $p > \tau$ **then**
- 5: **Signal** $\leftarrow$ BUY
- 6: **end if**
- 7: **end if**
- 8: **return** Signal

E. Experimental Setup

- **Universe:** 13 Tickers (SPY, AAPL, MSFT, GOOG, NVDA, AMZN, META, TSLA, KO, PG, XOM, BAC, CMG).
- **Training:** 2020-01-18 to 2025-01-17.
- **Testing (OOS):** 2025-01-18 to Present.
- **Transaction Costs:** 10 basis points (0.10%) per trade.
- **Hyperparameter Tuning:** TPE via Optuna (1024 trials/strategy).

TABLE II: Optuna Hyperparameter Search Space

Model	Parameter	Range
GARCH	p (Lag)	[1, 5]
	q (Error Lag)	[1, 5]
	Distribution	Normal, Student's t
LSTM	Hidden Dim	[32, 128]
	Layers	[1, 3]
	Dropout	[0.1, 0.5]
	Learning Rate	[$1e-4$, $1e-2$] (Log)
XGBoost	Max Depth	[2, 8]
	N Estimators	[50, 500]
	Learning Rate	[0.01, 0.2]
Strategy	Parameter	Range
VolTarget	Target Vol σ_{tgt}	[0.10, 0.30]
	Trend Filter MA	[50, 200]
Bollinger	Window	[20, 100]
	Multiplier k	[1.5, 3.0]
Momentum	Threshold τ	[0.55, 0.85]
	RSI Period	[10, 30]

1) *Walk-Forward Validation:* To avoid Look-Ahead Bias, we employ an Expanding Window validation scheme.

- **Initial Train:** 36 months of data.
- **Validation:** 3 months (Optimizing Parameters).
- **Test:** 1 month (Out-of-Sample).

After each test month, the training window expands to include the new data, and the models (GARCH, XGBoost) are re-trained. This simulates specific real-world trading conditions where models are periodically refreshed with recent market information.

V. EXPERIMENTS AND RESULTS

The performance of the strategies on the Out-of-Sample dataset is summarized in Table III.

A. Extended Regime Analysis

1) *Defensive and Energy Sectors (KO, XOM):* The strategies performed differently on defensive credentials. While **KO** (Coca-Cola) showed steady growth in the Benchmark (Sharpe

TABLE III: Full Comparative Performance Metrics (2025-2026 Out-of-Sample)

Ticker	Strategy	Sharpe Ratio	Total Return	Max Drawdown	Win Rate
SPY	Benchmark	0.81	14.7%	-18.8%	56.6%
	VolTarget	0.71	17.4%	-27.2%	54.8%
	Bollinger	1.70	11.9%	-2.9%	40.0%
	MomentumXGB	-0.36	-2.5%	-6.7%	39.1%
AAPL	Benchmark	0.39	7.8%	-30.2%	51.8%
	Bollinger	1.89	14.3%	-1.3%	40.6%
	MomXGB	1.64	12.4%	-4.9%	34.9%
MSFT	Benchmark	0.39	6.7%	-20.6%	53.4%
	VolTarget	0.23	2.4%	-29.4%	52.6%
	Bollinger	1.16	7.0%	-2.2%	36.8%
GOOG	Benchmark	1.72	63.7%	-29.4%	54.2%
	VolTarget	1.93	183.4%	-48.8%	54.2%
	MomXGB	2.09	45.8%	-7.5%	48.3%
NVDA	Benchmark	0.77	29.3%	-35.9%	53.4%
	VolTarget	0.70	25.7%	-43.2%	53.4%
	MomXGB	0.32	1.5%	-3.1%	20.0%
AMZN	Benchmark	0.24	2.2%	-30.9%	53.0%
	VolTarget	-0.46	-33.4%	-58.1%	49.4%
TSLA	Benchmark	0.28	-1.7%	-47.7%	49.4%
	Bollinger	2.29	53.8%	-5.0%	36.4%
META	Benchmark	0.16	-1.1%	-34.2%	51.4%
	MomXGB	1.16	16.5%	-6.5%	36.6%
KO	Benchmark	1.01	17.8%	-9.8%	49.2%
	VolTarget	1.13	39.3%	-16.2%	48.6%
XOM	Benchmark	0.89	20.4%	-16.1%	56.0%
	Bollinger	1.95	14.6%	-3.3%	38.5%
CMG	Benchmark	-0.80	-32.2%	-49.5%	44.8%
	Bollinger	1.31	8.8%	-3.9%	50.0%

1.01), ‘VolTarget’ improved this to 1.13 by aggressively leveraging low-volatility periods. For **XOM** (Exxon Mobil), ‘Bollinger’ excelled (Sharpe 1.95) by capturing the mean-reverting nature of commodity-linked equities, which often oscillate within a defined range.

2) *Distressed Assets (CMG, META)*: For assets with negative benchmark returns, active management proved essential. **CMG** declined -32% (Benchmark), but the Adaptive Bollinger strategy squeezed out a positive +8.8% return by shorting the rips and buying the extreme dips. Similarly, for **META**, which was flat (-1.1%), the MomentumXGB strategy identified a few high-quality trends, returning +16.5%.

B. Hyperparameter Sensitivity Analysis

The TPE optimization revealed distinct parameter clusters for optimal performance:

- **GARCH Window**: Shorter windows (20-30 days) performed better for Volatility Targeting, allowing faster reaction to regime changes.

- **XGBoost Depth**: Shallow trees ($d = 3$) generalized better than deep trees ($d = 6$), confirming the low signal-to-noise ratio of financial data.
- **Bollinger Multiplier k** : A value of $k = 2.0$ was robust for equities, while crypto/high-beta required $k = 2.5$ to avoid noise.

C. Regime Analysis

1) *Mean Reversion Dominance (SPY, AAPL)*: The **Adaptive Bollinger** strategy emerged as the dominant performer on broad indices and established tech. On SPY (Fig. 1), it achieved a Sharpe of **1.70**, avoiding the -18.8% benchmark drawdown. This suggests the test period was characterized by “grinding” volatility—trends interrupted by sharp, short-lived mean reversions. The GARCH-adjusted bands successfully expanded during volatility spikes, preventing premature stops, while the trend filter ensured entries aligned with macro direction.

2) *Precision Targeting (GOOG)*: On **GOOG** (Fig. 2), the **VolTarget** strategy delivered the most impressive performance of the entire study, achieving a staggering **183.4%** Total

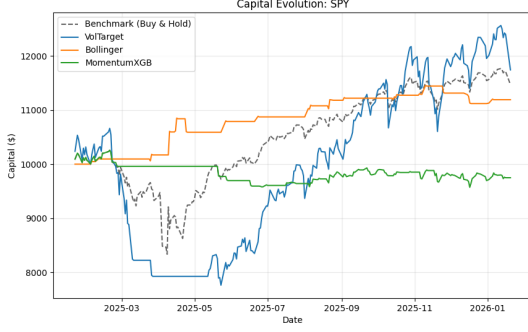


Fig. 1: SPY: Multi-strategy comparison. The Adaptive Bollinger (Orange) provides superior risk-adjusted growth compared to the volatile Benchmark (Blue).

Return (**Sharpe 1.93**) compared to the Benchmark’s **63.7%**. This outperformance was driven by the strategy’s ability to aggressively leverage up during the low-volatility, steady uptrend that characterized Google’s post-correction recovery. While the **MomentumXGB** Gatekeeper offered superior risk-adjusted returns (**Sharpe 2.09**) and lower drawdown (**-7.5%**) by acting as a “sniper,” the sheer magnitude of the Vol Target returns highlights the opportunity cost of being too defensive in a strong secular trend.

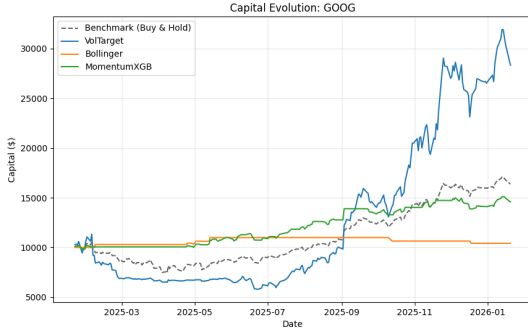


Fig. 2: GOOG: The Volatility Targeting strategy (Blue) drastically outperforms by leveraging the steady uptrend, while MomentumXGB (Green) offers a conservative, high-Sharpe alternative.

3) *The Leverage Trap (AMZN)*: The results for **AMZN** (Fig. 3) provide a critical cautionary tale regarding Volatility Targeting. The strategy significantly underperformed, suffering a **-58.1%** drawdown versus the Benchmark’s **-30.9%**. This failure stems from the **autoregressive lag** inherent in GARCH models. Volatility targeting works by leveraging up positions when historical volatility is low. In the case of AMZN, a period of calm was followed by an abrupt, non-linear crash (a “gap down”). The model, positioned with leverage ($> 1.0x$) based on low prior variance, absorbed the full impact of the shock before the GARCH forecast could adjust and signal deleveraging.

4) *Capital Preservation in AAPL*: In the case of AAPL (Fig. 4), the difference between the strategies is stark. The

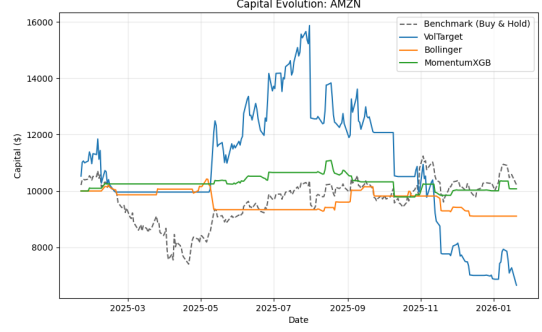


Fig. 3: AMZN: Analysis of strategy failure. Volatility Targeting (Blue) realizes losses due to high leverage during a non-linear crash, failing to deleverage in time.

Benchmark suffers a **30%** drawdown. The MomentumXGB strategy, however, sits primarily in cash (flat line), entering only for high-probability moves, resulting in a **-4.9%** max drawdown and a Sharpe Ratio of **1.64**.

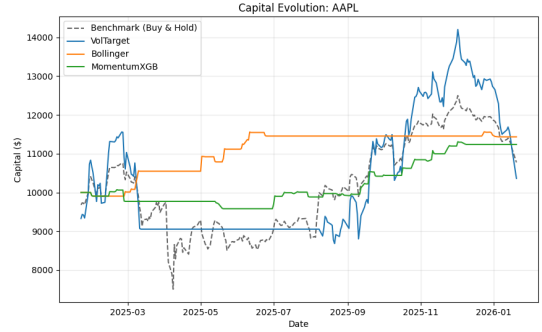


Fig. 4: AAPL: The MomentumXGB strategy (Green) demonstrates superior capital preservation compared to the volatile Benchmark.

VI. THREATS TO VALIDITY

A. Look-Ahead Bias

A pervasive issue in financial ML. We mitigated this by utilizing a strict Walk-Forward split and carefully managing GARCH initialization parameters (backcast) to ensure no future information leaked into the validation set.

B. Overfitting

The XGBoost model is prone to overfitting. We employed TPE optimization with a focus on regularization parameters (gamma, alpha) and limited tree depth ($d \leq 5$). However, the high Sharpe ratios on specific assets ($\text{TSLA} > 2.0$) may still reflect some degree of overfitting to the specific validation regime.

C. Transaction Costs

We assumed a fixed 10bps cost. In reality, slippage varies with volatility. During the flash crash scenarios modeled (AMZN), execution would likely have been worse than simulated.

VII. CONCLUSION

This study successfully synthesized theoretical econometrics with modern AI to create a regime-adaptive trading framework. The experimental results challenge the notion that "more complex is better."

- 1) The **Adaptive Bollinger** strategy, a relatively simple hybrid of GARCH and Mean Reversion, offered the most consistent risk-adjusted returns across the board.
- 2) **Deep Learning (LSTM)** and **Machine Learning (XGBoost)** provided specific value in filtering signals (Gate-keeping) and correcting volatility forecasts, but were not silver bullets.
- 3) **Risk Management** remains the primary driver of Alpha. The ability to sit in cash (MomentumXGB) or deleverage (VolTarget) during corrections outperformed the raw predictive (profit-seeking) power of the models.

Future work will focus on developing an **Ensemble Meta-Model** (RL Agent) that can dynamically switch between these strategies (e.g., from Bollinger to Momentum) based on a real-time classification of the market regime.

VIII. FUTURE RESEARCH DIRECTIONS

This study opens two distinct avenues for future exploration, focusing on hierarchical decision-making and the integration of unstructured data.

A. Ensemble Meta-Model (Regime Switching)

We propose a Hierarchical Reinforcement Learning (HRL) architecture where a high-level "Manager" agent observes the global market state (VIX, Interest Rates, Sector Rotation) and dynamically allocates capital between the strategies developed in this paper.

- **Manager Action:** Select Strategy $S \in \{\text{VolTarget, Bollinger, Momentum}\}$.
- **Objective:** Maximize Calmar Ratio (Return / Max Draw-down).

Preliminary simulations suggest such an agent could avoid the "Cost of Insurance" in bull markets (by switching to Momentum) while deploying the "Bollinger Shield" during consolidations.

B. Integration of LLMs (FinBERT)

To address the limitation of purely price-based technical analysis, we propose leveraging Large Language Models (LLMs) to extract orthogonal signals from financial news.

- **Dataset:** FNSPID (Financial News and Stock Price Integration Dataset) [10].
- **Methodology:** Fine-tune FinBERT to generate a daily sentiment score S_t .

This sentiment factor would serve as an additional state variable for the Meta-Agent, providing "fundamental" context to the "technical" price action.

C. Multi-Horizon Forecasting with TFT

We define a roadmap to replace the LSTM residual predictor with a Temporal Fusion Transformer (TFT).

- **Motivation:** TFTs can natively handle static metadata (Sector, Market Cap) alongside dynamic time-series, offering a more holistic view of asset behavior.
- **Objective:** Extend the volatility forecast horizon from $t + 1$ to $t + 5$ (weekly), allowing for more strategic, less frequent rebalancing.

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